ASSIGNMENT#2

BT201: Ecology, Biodiversity & Evolution-II

Spring 2021

Total Marks: 10
Due date: 28-07-2021

VU Medical Zone Effort By Amaan Khan

Question No. 1

Briefly discuss the contribution of following scientists: (4X2=8)

a) Turesson

Turesson is remembered by botanists as a pioneer in the use of experimental methods to detect genetic differences between races of plant species that adapt them to different habitats. He coined two terms, ecotype and agamospecies, that are currently used widely by ecological geneticists and evolutionary botanists.

b) Kerner

Kerner von Marilaun (1831-1898), who is known for documenting the flora of the Austro-Hungarian empire. In addition to compiling a list of the plant species that occurred within the empire's sprawling borders, Kerner also analyzed plant distributions at various scales across both space and time, demonstrating that vegetation is not only a reflection of the contemporary climate, but also that changes in species composition are in response to changes in climate. Kerner determined that the boundaries for vegetation types and growth "resulted from the same effect: the limited duration of the growing season in each climate

c) Hardy-Weinberg

Hardy-Weinberg law, an algebraic equation that describes the genetic equilibrium within a population. It was discovered independently in 1908 by Wilhelm Weinberg, a German physician, and Godfrey Harold Hardy, a British mathematician.In population genetics, the Hardy–Weinberg principle, also known as the Hardy–Weinberg equilibrium, model, theorem, or law, states that

allele and genotype frequencies in a population will remain constant from generation to generation in the absence of other evolutionary influences.

d) Frankham

Population fragments with limited gene flow can become genetically differentiated through random effects of genetic drift and fixation of alleles due to inbreeding, resulting in decreased fitness. Genetic factors are important to consider when establishing new populations as inbreeding and outbreeding depression, genetic erosion, and loss of genetic diversity can have significant impacts on time to extinction. The role of genetic factors in extinctions of wild populations has been controversial, but evidence now shows that they make important contributions to extinction risk. Inbreeding has been shown to cause extinctions of wild populations, computer projections indicate that inbreeding depression has important effects on extinction risk, and most threatened species show signs of genetic deterioration.

Question No. 2

What is the temperature range of reef-building corals? (2)

Reef-building corals cannot tolerate water temperatures below 64° Fahrenheit (18° Celsius). Many grow optimally in water temperatures between 73° and 84° Fahrenheit (23°–29°Celsius)

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